1 Integral Curves

Integral Curves

Definition 1.1. : For a vector field V on a manifold M, **an integral curve of** V is a smooth curve $\gamma : I(\subseteq \mathbb{R}) \to M$ whose velocity at each point is equal to the value of V at that point, i.e.,

$$\gamma'(t) = V_{\gamma(t)} \quad (\forall t \in I)$$

If $0 \in I$, then the point $\gamma(0)$ is called the **starting point of** γ.

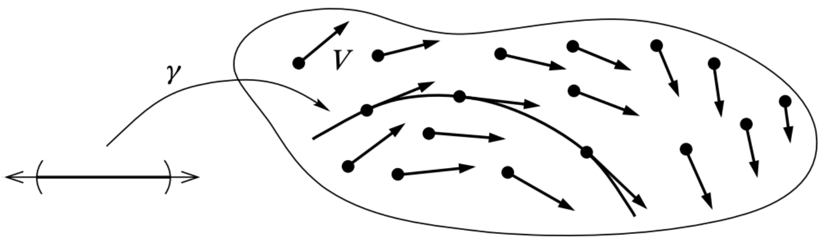

图 1: An integral curve of a vector field

Proposition 1.1. : *Let V be a smooth vector field on a smooth manifold M. For each $p \in M$, there exists $\varepsilon > 0$ and a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ that is an integral curve of V starting at p.*

Proof: For any $t \in I$, take a local coordinate $(U; x^1, x^2, \ldots, x^n)$ of $\gamma(t)$, we can write γ as

$$\gamma(t) = (\gamma^1(t), \gamma^2(t), \ldots, \gamma^n(t))$$

then the condition $\gamma'(t) = V_{\gamma(t)}$ can be written as

$$\dot{\gamma}^i(t) \frac{\partial}{\partial x^i}\bigg|_{\gamma(t)} = V^i(\gamma(t)) \frac{\partial}{\partial x^i}\bigg|_{\gamma(t)} \quad (i = 1, 2, \ldots, n),$$

which reduces to a system of ordinary differential equations

$$\dot{\gamma}^i(t) = V^i(\gamma(t)) \quad (i = 1, 2, \ldots, n)$$

with initial condition

$$\gamma^i(0) = x^i(p) \quad (i = 1, 2, \ldots, n)$$

By the theory of ODE, we know that the solutions of such equations exists and is unique, which implies that the propostion holds. □

The following two lemmas show how affine reparametrizations affect integral curves.

Rescaling Lemma

Lemma 1.1. : *Let V be a smooth vector field on a smooth manifold M. Let $J \subseteq \mathbb{R}$ be an interval and let $\gamma : J \to M$ be an integral curve of V. For any $a \in \mathbb{R}$, the curve $\tilde{\gamma} : \tilde{J} \to M$ defined by $\tilde{\gamma}(t) := \gamma(at)$ is an integral curve of the vector field aV, where $\tilde{J} : \{t \in \mathbb{R} | \ at \in J\}$.*

Proof: For any $t_0 \in \tilde{J}$, take any $f \in C^\infty_{\tilde{\gamma}(t_0)}(M)$, then we have

$$\tilde{\gamma}'(t_0)(f) = d\tilde{\gamma}_{t_0}(\frac{d}{dt}\Big|_{t_0})(f) = \frac{d}{dt}\Big|_{t=t_0}(f \circ \tilde{\gamma})(t) = \frac{d}{dt}\Big|_{t=t_0}(f \circ \gamma)(at)$$

$$= a(f \circ \gamma)'(at_0) = a\gamma'(at_0)(f) = aV_{\tilde{\gamma}(t_0)}(f),$$

which means that $\tilde{\gamma}'(t) = aV_{\tilde{\gamma}(t)}$ for any $t \in \tilde{J}$. □

Translation Lemma

Lemma 1.2. : *Let V, M, J and γ be as in the preceding lemma. For any $b \in \mathbb{R}$, the curve $\hat{\gamma} : \hat{J} \to M$ defined by $\hat{\gamma}(t) := \gamma(t+b)$ is also an integral curve of V, where $\hat{J} = \{t \in \mathbb{R} | \ t+b \in J\}$.*

Proof: Similar to the proof in the preceding lemma. □

Recall: For a smooth map $F : M \to N$ between two manifolds and vector fields X, Y on M, N respectively, we say that X, Y are F-related if

$$Y_{F(p)} = dF_p(X_p) \quad (\forall p \in M).$$

Naturality of Integral Curves

Proposition 1.2. : *Suppose M, N are two smooth manifolds and $F : M \to N$ is a smooth map. Let X be a smooth vector field on M and Y a smooth vector field on N. Then X, Y are F-related $\iff$ F takes integral curves of X to integral curves of Y, meaning that for each integral curve γ of X, $F \circ \gamma$ is an integral curve of Y.*

Proof: ($\Rightarrow$): Let $\gamma : J \to M$ be an integral curve if X. If we define $\sigma : J \to N$ by $\sigma = F \circ \gamma$, then

$$\sigma'(t) = (F \circ \gamma)'(t) = dF_{\gamma(t)}(\gamma'(t)) = dF_{\gamma(t)}(X_\gamma(t)) = Y_{F(\gamma(t))} = Y_\sigma(t)$$

which means that $\sigma = F \circ \gamma$ is an integral curve of Y;

($\Leftarrow$): For any $p \in M$, by Proposition 1.1, we know that there exists an integral curve $\gamma : (-\varepsilon, \varepsilon) \to M$ starting at $p \in M$. Since $F \circ \gamma$ is an integral curve of Y starting at $F(p) \in N$, we have

$$Y_{F(p)} = (F \circ \gamma)'(0) = dF_p(\gamma'(0)) = dF_p(X_p)$$

which shows that X, Y are F-related. □

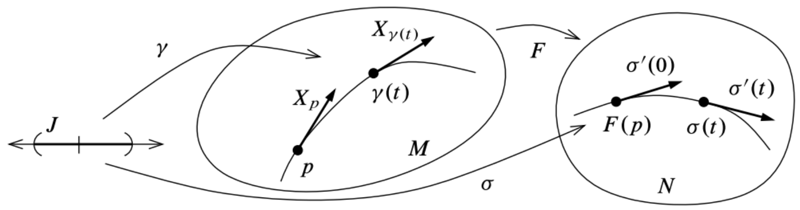

图 2: Flows of F-related vector fields

2 Flows

Let M be a smooth manifold and V be a smooth vector field on M, suppose for $\forall p \in M$, V has a unique integral curve starting at p and defined for all $t \in \mathbb{R}$, which we denoted by $\theta^{(p)} : \mathbb{R} \to M$. (It may not always be the case that every integral curve is defined for all t, but for purpose of illustration let us assume so for the time being.)

For each $t \in \mathbb{R}$, we define a map

$$\theta_t : M \to M$$

$$p \mapsto \theta_t(p) := \theta^{(p)}(t)$$

since we are assuming uniqueness of integral curve, the translation lemma implies that $t \mapsto \theta^{(p)}(t+s)$ is an integral curve of V startinig at $q \triangleq \theta^{(p)}(s)$. If we translate this into a statement about θ_t, it becomes

$$\theta_t \circ \theta_s(p) = \theta_{t+s}(p)$$

together with the equation $\theta_0(p) = \theta^{(p)}(0) = p$. This implies that the map

$$\theta : \mathbb{R} \times M \to M$$

$$(t, p) \mapsto \theta_t(p) = \theta^{(p)}(t)$$

is actually an action of the additive group $\mathbb{R}$ on M.

Motivited by these obeservations, we have the following defition about global flow.

Global Flow

Definition 2.1. : A **global flow** on a smooth manifold M is a continuous left $\mathbb{R}$-action ib M, which means a continuous map

$$\theta : \mathbb{R} \times M \to M$$

satisfying the following property:
(1) $\theta(0, p) = p \quad (\forall p \in M)$;
(2) $\theta(t, \theta(s, p)) = \theta(t + s, p) \quad (\forall t, s \in \mathbb{R}, \forall p \in M)$.

Remark: Given a global flow θ on M, we define two collections of maps as follows:

- For each $t \in \mathbb{R}$, define a continuous map $\theta_t : M \to M$ by

$$\theta_t(p) := \theta(t, p).$$

then the defining properties (1) and (2) are equivalent to the **group laws**

$$\theta_t \circ \theta_s = \theta_t + s, \quad \theta_0 = \mathrm{Id}_M.$$

- For $\forall p \in M$, define a curve

$$\theta^{(p)} : \mathbb{R} \to M$$
$$t \mapsto \theta^{(p)}(t) := \theta(t, p)$$

then we can see that the image of this curve is precisely the orbit of p under the group action.

If $\theta : \mathbb{R} \times M \to M$ is a smooth global flow, for any $p \in M$, we define a tangent vector $V_p \in T_pM$ by

$$V_p := (\theta^{(p)})'(0)$$

then the assignment $p \mapsto V_p$ is a vector field on M, which is called the **infinitesimal generator of** θ.

The following propostion shows that every smooth gloabl flow is derived from the integral curves of some smooth vector field in precisely the way we described above.

Proposition 2.1. *: Let $\theta : \mathbb{R} \times M \to M$ be a smooth global flow on a smooth manifold M. The infinitesimal generator V of θ is a smooth vector field on M, and each curve $\theta^{(p)}$ is an integral curve of V.*

Proof: Recall that a vector field V on M is smooth $\iff$ for any $f \in C^\infty(M)$, $V(f)$ is a smooth function on M.

Take $\forall f \in C^\infty(M), \forall p \in M$, we have

$$(V(f))(p) = V_p(f) = ((\theta^{(p)})'(0))(f) = \left.\frac{d}{dt}\right|_{t=0} f(\theta^{(p)}(t)) = \left.\frac{\partial}{\partial t}\right|_{(0,p)} f(\theta(t,p))$$

because $f(\theta(t,p))$ is a smooth function of (t,p) by composition, so is its partial derivative w.r.t. t, thus $V(f) \in C^\infty(M)$, hence $V \in \mathfrak{X}(M)$;

For any $t_0 \in \mathbb{R}$, set $q \triangleq \theta^{(p)}(t_0) = \theta_{t_0}(p)$, by the group law, we get

$$\theta^{(q)}(t) = \theta_t(q) = \theta_t(\theta_{t_0}(p)) = \theta_{t+t_0}(p) = \theta^{(p)}(t + t_0)$$

therefore, for any $f \in C^\infty(M)$, we have

$$V_q(f) = ((\theta^{(q)})'(0))(f) = \left.\frac{d}{dt}\right|_{t=0} f(\theta^{(q)}(t)) = \left.\frac{d}{dt}\right|_{t=0} f(\theta^{(p)}(t + t_0)) = ((\theta^{(p)})'(t_0))(f),$$

which means that $V_q = V_{\theta^{(p)}(t)} = (\theta^{(p)})'(t)$ for all $t \in \mathbb{R}$. $\square$

Remark: By this propostion we see that every smooth global flow gives rise to a smooth vector field whose integral curves are precisely the curve defined by the flow.

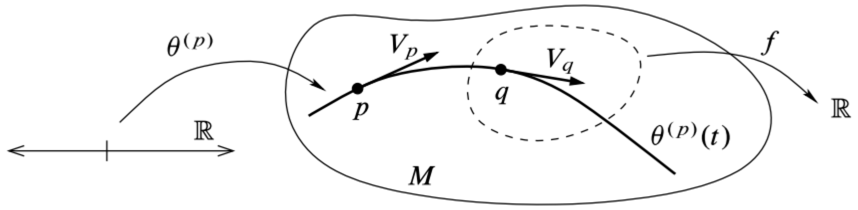

图 3: The infinitesimal generator of a global flow

Flow Domain

Definition 2.2. : Let M be a smooth manifold, a **flow domain** for M is an open subset $\mathcal{D} \subseteq \mathbb{R} \times M$ with the property that for each $p \in M$, the set $\mathcal{D}^{(p)} := \{t \in \mathbb{R} |\ (t, p) \in \mathcal{D}\}$ is an open interval containing 0.

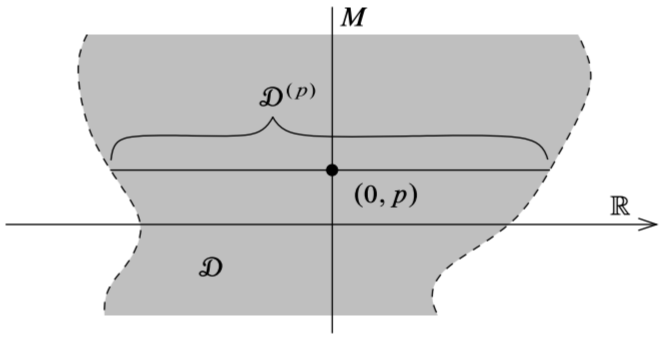

图 4: A flow domain

Flow

Definition 2.3. : A **flow** on M is a continuous map $\theta : \mathcal{D} \to M$, where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain, that satiefies the following group laws: for all $p \in M$,

$$\theta(0, p) = p,$$

and for all $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{(\theta(s,p))}$ such that $s + t \in \mathcal{D}^{(p)}$,

$$\theta(t, \theta(s, p)) = \theta(t + s, p).$$

Remark: We sometimes call θ a **local flow** to distinguish it from a global flow as defined earlier. The unwiedly term **local one-parameter group action** is also used.

Remark: If θ is a flow on M, we define $\theta_t(p) = \theta^{(p)}(t) := \theta(t, p)$ whenever $(t, p) \in \mathcal{D}$, just as for a global flow. For each $t \in \mathbb{R}$, we also define

$$M_t := \{p \in M |\ (t, p) \in \mathcal{D}\}$$

so that

$$p \in M \Leftrightarrow t \in \mathcal{D}^{(p)} \Leftrightarrow (t, p) \in \mathcal{D}.$$

Remark: If θ is a smooth flow on M, the **infinitesimal generator of** θ is a vector field V defined by $V_p := (\theta^p)'(0)$.

Proposition 2.2. *: If $\theta : \mathcal{D} \to M$ is a smooth flow, then the infinitesimal generator V of θ is a smooth vector field and each curve $\theta^{(p)}$ is an integral curve of V.*

Proof: In the proof that V is smooth, just note that for any $p_0 \in M$, $\theta(t,p)$ is defined and smooth for all (t,p) sufficiently close to $(0,p_0)$ since $\mathcal{D}$ is open. In the proof that $\theta^{(p)}$ is an inetgral curve, it is the same as that in Proposition 2.1. □

A **maximal integral curve** is one that cannot be extended to an integral curve on any larger open interval and a **maximal flow** is a flow that admits no extension to a flow on a larger flow domain.

Fundamental Theorem on Flows

Theorem 2.1. *: Let V be a smooth vector field on a smooth maifold M. Then there is a unique maximal flow $\theta : \mathcal{D} \to M$ whose infinitesimal generator is V. This flow has the following properties:*

(1) For each $p \in M$, the curve $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ is the unique maximal integral curve of V starting at p.

(2) If $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))}$ is the interval $\mathcal{D}^{(p)} - s = \{t - s |\ t \in \mathcal{D}^{(p)}\}$.

(3) For each $t \in \mathbb{R}$, the set M_t is open in M and $\theta_t : M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}.

Proof: (1) By Proposition 1.1, there exists an intergal curve starting at each point $p \in M$. Suppose $\gamma, \tilde{\gamma} : J \to M$ are two integral curves of V defined on the same interval J such that $\gamma(t_0) = \tilde{\gamma}(t_0)$ for some $t_0 \in J$. Denote $S := \{t \in J |\ \gamma(t) = \tilde{\gamma}(t)\}$, so $S \neq \phi$ since $t_0 \in S$. By continuity, S is closed. On the other hand, suppose $t_1 \in S$, then in a local coordinate about $p = \gamma(t_1)$, γ and $\tilde{\gamma}$ are both solutions to the same ODE with the same initial condition that $p = \gamma(t_1) = \tilde{\gamma}(t_1)$. By the uniqueness from ODE theory, we know that $\gamma \equiv \tilde{\gamma}$ on an interval containing t_1, which implies S is open in J. Since J is connected and S is both open and closed in J and $S \neq \phi$, we get $S = J$, thus $\gamma \equiv \tilde{\gamma}$ on all of J, i.e., any two integral curves that agree at one point agree on their common domain. Now, for each $p \in M$, let $\mathcal{D}^{(p)}$ be the union of all open intervals $J \subseteq \mathbb{R}$ containing 0 on which an integral curve starting at p is defined. Define $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ by $\theta^{(p)}(t) := \gamma(t)$, where γ is any integral curve starting at p and defined on an open interval containing 0 and t. By the argument before, $\theta^{(p)}$ is well-defined and is obviously the unique maximal integral curve starting at p;

(2) Let $\mathcal{D} = \{(t,p) \in \mathbb{R} \times M |\ t \in \mathcal{D}^{(p)}\}$, define $\theta : \mathcal{D} \to M$ by $\theta(t,p) := \theta^{(p)}(t)$, we also write $\theta_t(p) := \theta(t,p)$ as usual. By (1), we know that $\theta^{(p)}$ is the unique maximal integral curve of V starting at $p \in M$. To verify the group laws, fix any $p \in M$ and $s \in \mathcal{D}^{(p)}$, denote $q \triangleq \theta(s,p) = \theta^{(p)}(s)$. The curve $\gamma : \mathcal{D}^{(p)} - s \to M$ defined by $\gamma(t) := \theta^{(p)}(t+s)$ starts at q and the translation lemma shows that γ is an integral curve of V. By the uniqueness from ODE solutions, γ agrees with $\theta^{(q)}$ on their common domain, from which we get $\gamma(t) = \theta^{(q)}(t)$, i.e., $\theta(t+s,p) = \theta(t, \theta(s,p))$ and

$\theta(0,q)=q$ is immediate. By the maximality of $\theta^{(q)}$, the domain of γ cannot be larger than $\mathcal{D}^{(q)}$, hence $\mathcal{D}^{(p)}-s\subseteq\mathcal{D}^{(q)}$. Since $0\in\mathcal{D}^{(p)}$, so $-s\in\mathcal{D}^{(q)}$ and the group law implies that $\theta^{(q)}(-s)=p$. Applying the same argument with $(-s,q)$ in place of (s,p), we can know get $\mathcal{D}^{(q)}+s\subseteq\mathcal{D}^{(p)}$;

(3) First we show that $\mathcal{D}$ is open in $\mathbb{R}\times M$ (so it is a flow domain) and that $\theta:\mathcal{D}\to M$ is smooth:

Define a subset $W\subseteq\mathcal{D}$ as the set of all $(t,p)\in D$ such that θ is defined and smooth on a product neighborhood of (t,p) of the form $J\times U\subseteq\mathcal{D}$, where $J\subseteq\mathbb{R}$ is an open interval containing 0 and t and $U\subseteq M$ is an open neighborhood of p. Then W is open in $\mathbb{R}\times M$ and the restriction of θ to W is smooth, so it suffices to show that $\mathcal{D}=W$. Suppose this is not the case, then there exists some point $(\tau,p_0)\in\mathcal{D}\setminus W$, for simplicity, assume $\tau>0$ (the argument that $\tau<0$ is similar).

Let $t_0:=\inf\{t\in\mathbb{R}|\ (t,p_0)\notin W\}$, by the ODE theorem (applied in smooth coordinate around p_0), θ is defined and smooth in some product neighborhood of $(0,p_0)$, hence $t_0>0$. Since $t_0\le\tau$ and $\mathcal{D}^{(p_0)}$ is an open interval containing 0 and τ, it follows that $t_0\in\mathcal{D}^{(p_0)}$. Let $q_0\triangleq\theta^{(p_0)}(t_0)$, by the ODE theorem again, there exists $\varepsilon>0$ and a neighborhood U_0 of q_0 such that $(-\varepsilon,\varepsilon)\times U_0\subseteq W$. We use the group law to show that θ extends smoothly to a neighborhood of (t_0,p_0), which contradicts to the choice of t_0:

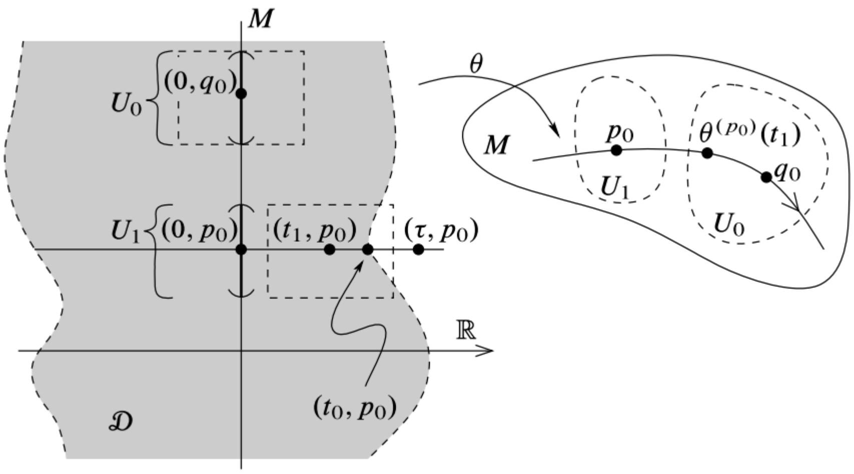

图 5: Proof that $\mathcal{D}$ is open

Choose some $t_1<t_0$ such that $t_1+\varepsilon>t_0$ and $\theta^{(p_0)}(t_1)\in U_0$. Since $t_1<t_0$, we have $(t_1,p_0)\in W$ and so there is a product neighborhood $(t_1-\delta,t_1+\delta)\times U_1\subseteq W$. By the definition of W, by the definition of W, this implies that θ is defined and smooth on $[0,t_1+\delta)\times U_1$. Because $\theta(t_1,p_0)\in U_0$, we can choose U_1 small enough that θ maps $\{t_1\}\times U_1$ into U_0. Define $\tilde\theta:[0,t_1+\varepsilon)\times U_1\to M$ by

$$\tilde\theta(t,p):=\begin{cases}\theta_t(p), & p\in U_1, 0\le t<t_1\\ \theta_{t-t_1}\circ\theta_{t_1}(p), & p\in U_1, t_1-\varepsilon<t<t_1+\varepsilon\end{cases}$$

the group law for θ guarantees that these definitions agree where they overlap, and our choice of U_1, t_1 and ε ensure that this defines a smooth map. By the translation lemma, each map $t\mapsto\tilde\theta(t,p)$ is an integral curve of V, so $\tilde\theta$ is a smooth extension of θ to a neighborhood of (t_0,p_0), contradicting to our choice of t_0. Thus we get $W=\mathcal{D}$.

Now, we prove (3): Since $\mathcal{D}$ is open, it's clear that M_t is open in M. By the conclusion of (2), we get

$$p \in M_t \Rightarrow t \in \mathcal{D}^{(p)} \Rightarrow \mathcal{D}^{(\theta_t(p))} = \mathcal{D}^{(p)} - t \Rightarrow -t \in \mathcal{D}^{(\theta_t(p))} \Rightarrow \theta_t(p) \in M_{-t}$$

which means that for $\theta_t : M_t \to M_{-t}$. Since $\theta_{-t} \circ \theta_t = \mathrm{Id}_M$, thus $\theta_{-t} \circ \theta_t$ is equal to the identity map on M_t. $\square$

Remark: The flow whose existence and uniqueness are asserted in the fundamental theorem is called the **flow generated by** V, or the **flow of** V.

The naturality of integral curves translates into the following naturality statement for flows.

Naturality of Flows

Proposition 2.3. : *Suppose M, N are smooth manifolds, $F : M \to N$ is a smooth map, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. Let θ be the flow generated by X and η the flow generated by Y. If X, Y are F-related, then for each $t \in \mathbb{R}$, $F(M_t) \subseteq N_t$ and $\eta_t \circ F = F \circ \theta_t$, i.e., we have the follwing diagram*

$$\begin{tikzcd} M_t \arrow[r, "F"] \arrow[d, "\theta_t"] & N_t \arrow[d, "\eta_t"] \\ M_{-t} \arrow[r, "F"'] & N_{-t} \end{tikzcd}$$

Proof: By Proposition 1.2, for any $p \in M$, the curve $F \circ \theta^{(p)}$ is an integral curve of Y starting at $F(p)$. By the uniqueness of integral curves, the maximal integral curve $\eta^{(F(p))}$ must be defined at least on the interval $\mathcal{D}^{(p)}$ and $F \circ \theta^{(p)} = \eta^{(F(p))}$ on that interval, which means that

$$p \in M_t \Rightarrow t \in \mathcal{D}^{(p)} \Rightarrow t \in \mathcal{D}^{(F(p))} \Rightarrow F(p) \in N_t,$$

hence $F(M_t) \subseteq N_t$ and $F(\theta^{(p)}(t)) = \eta^{(F(p))}(t)$ $(\forall t \in \mathcal{D}^{(p)})$, i.e.,

$$\eta_t \circ F(p) = F \circ \theta_t(p) \ (\forall p \in M_t),$$

which means that the proposition hlods. $\square$

Diffeomorphism Invariance of Flows

Corollary 2.1. : *Let $F : M \to N$ be a diffeomorphism. If $X \in \mathfrak{X}(M)$ and θ is the flow generated by X, then the flow of F_*X is $\eta_t = F \circ \theta_t \circ F^{-1}$.*

Proof: Just note that the pushfoward F_*X and X are F-related. $\square$

Since not every smooth vector field generates a global flow, we have the following definition.

Complete Vector Fields

Definition 2.4. : A smooth vector field $V \in \mathfrak{X}(M)$ is called **complete** if it generates a global flow, or equivalently if each of its maximal integral curves is defined for all $t \in \mathbb{R}$.

Uniform Time Lemma

Lemma 2.1. *: Let $V \in \mathfrak{X}(M)$ and let θ be the flow generated by V. Suppose there is a positive number ε such that for every $p \in M$, the domain of $\theta^{(p)}$ contains $(-\varepsilon, \varepsilon)$. Then V is complete.*

Proof: (By contradiction) Suppse the domain $\mathcal{D}^{(p)}$ of $\theta^{(p)}$ is bounded above. (A similar proof works if it is bounded below). Denote $b := \sup \mathcal{D}^{(p)}$, choose a positive number t_0 such that $b - \varepsilon < t_0 < b$, denote $q \triangleq \theta^{(p)}(t_0)$. By the hypothesis, $\theta^{(q)}(t)$ is defined at least for $t \in (-\varepsilon, \varepsilon)$. Define a curve $\gamma : (-\varepsilon, t_0 + \varepsilon) \to M$ by

$$\gamma(t) := \begin{cases} \theta^{(p)}(t), & -\varepsilon < t < b \\ \theta^{(q)}(t - t_0), & t_0 - \varepsilon < t < t_0 + \varepsilon \end{cases}$$

It's clear that γ is well defined since $\theta^{(q)}(t - t_0) = \theta_{t-t_0}(q) = \theta_{t-t_0} \circ \theta_{t_0}(p) = \theta_t(p) = \theta^{(p)}(t)$. By the translation lemma, γ is an integral curve of V starting at $p = \gamma(0)$. However, $t_0 + \varepsilon > b$, which contradicts to the definition of b. $\square$

Theorem 2.2. *: Every compactly supported smooth vector field on a smooth manifold is complete.*

Proof: Let $V \in \mathfrak{X}(M)$ be a compactly supported vector field on M, $K := \operatorname{supp} V$. Then for any $p \in K$, there exists a positive number ε_p such that the flow generated by V is defined at least on $(-\varepsilon_p, \varepsilon_p) \times U_p$. By the compactness of K, finitely many such sets $U_{p_1}, \ldots, U_{p_k}$ covers K. Take $\varepsilon := \min\{\varepsilon_{p_1}, \ldots, \varepsilon_{p_k}\}$, thus every maximal intgral curve starting in K is defined at least on $(-\varepsilon, \varepsilon)$. Since $V \equiv 0$ outside of K , every integral curve starting in $M \setminus K$ is constant and thus can be defined on all of $\mathbb{R}$. By the uniform time lemma, the conclusion holds. $\square$

Immediately, we have the following corollary.

Corollary 2.2. *: On a compact smooth manifold, every smooth vector field is complete.*